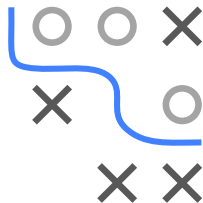


Optimization in Machine Learning

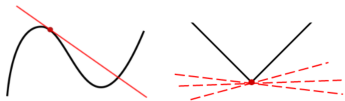
Foundations

Differentiation and Derivatives



Learning goals

- Uni- & multivariate differentiation
- Gradient, partial derivatives
- Jacobian matrix
- Hessian matrix
- Lipschitz continuity

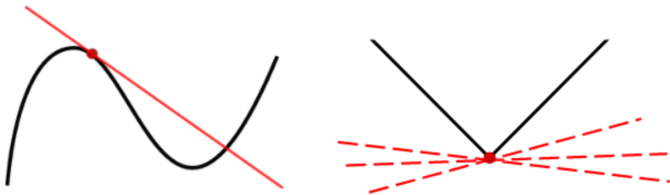
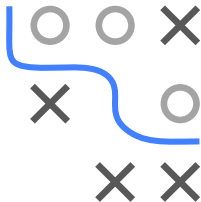


UNIVARIATE DIFFERENTIABILITY

Definition: A function $f : \mathcal{S} \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is said to be **differentiable** for each inner point $x \in \mathcal{S}$ if the following limit exists:

$$f'(x) := \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

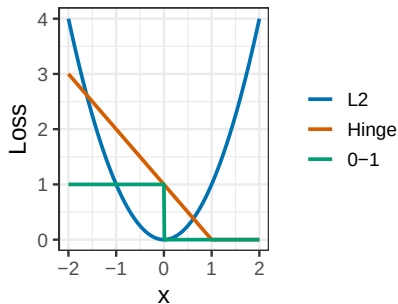
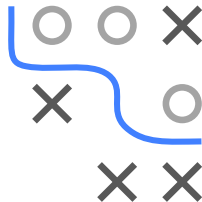
Intuitively: f can be approximated locally by a lin. fun. with slope $m = f'(x)$.



Left: Function is differentiable everywhere. **Right:** Not differentiable at the red point.

SMOOTH VS. NON-SMOOTH

- **Smoothness** of a function $f : \mathcal{S} \rightarrow \mathbb{R}$ is measured by the number of its continuous derivatives
- $\mathcal{C}^k$ is class of k -times continuously differentiable functions ($f \in \mathcal{C}^k$ means $f^{(k)}$ exists and is continuous)
- In this lecture, we call f “smooth”, if at least $f \in \mathcal{C}^1$



f_1 is smooth, f_2 is continuous but not differentiable, and f_3 is non-continuous.

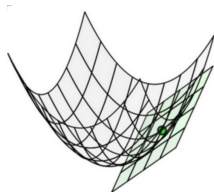
MULTIVARIATE DIFFERENTIABILITY

Definition: For a function $f : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ of d variables $x_1, \dots, x_d$, **partial derivatives** are defined as

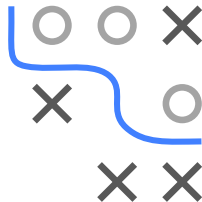
$$\frac{\partial f}{\partial x_1} = \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_d) - f(\mathbf{x})}{h}$$

$\vdots$

$$\frac{\partial f}{\partial x_d} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{d-1}, x_d + h) - f(\mathbf{x})}{h}$$



Geometrically: Similarly to the 1D case, the vector of partial derivatives can be used to determine a tangent hyperplane. [▶ Click for source](#)



DIRECTIONAL DERIVATIVE

The **directional derivative** tells how fast $f : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ is changing w.r.t. an arbitrary direction $\mathbf{v}$:

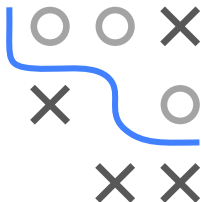
$$D_{\mathbf{v}}f(\mathbf{x}) := \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h} = \nabla f(\mathbf{x})\mathbf{v}$$

Example: The directional derivative for $\mathbf{v} = (1, 1)$ is:

$$\begin{aligned} D_{\mathbf{v}}f(\mathbf{x}) &= \nabla f(\mathbf{x}) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} \end{aligned}$$

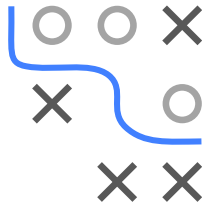
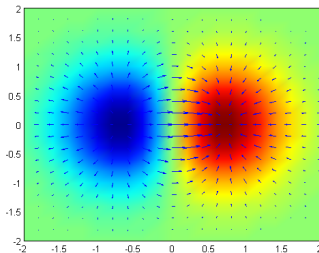
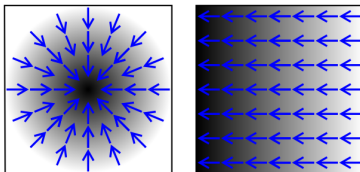
NB: Some people require that $\|\mathbf{v}\| = 1$. Then, we can identify $D_{\mathbf{v}}f(\mathbf{x})$ with the instantaneous rate of change in direction $\mathbf{v}$, i.e.

$\lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h}$ – and in our example we would have to divide by $\sqrt{2}$.



IMPORTANT PROPERTIES OF THE GRADIENT

- **Orthogonal** to level curves/surfaces of a function
- Points in direction of **greatest increase** of f



A 3x3 grid with a blue path starting at the top-left corner (0,0) and ending at the middle-right cell (1,2). The path consists of the following cells: (0,0), (0,1), (0,2), (1,2). Obstacles (X) are located at (0,2), (1,0), (2,0), and (2,1). Free cells (O) are located at (0,1), (1,1), and (2,2).

- $$\left\langle \nabla_{\mathbf{x}} f(\mathbf{x}_0), \frac{d\mathbf{x}}{dt}(t_0) \right\rangle = \nabla_{\mathbf{x}} f(\mathbf{x}(t_0)) \frac{d\mathbf{x}}{dt}(t_0) = \frac{d(f \circ \mathbf{x})}{dt}(t_0)$$

- Since $f \circ \mathbf{x}$ is a constant function in t , the derivative will always equal zero and so $\frac{d(f \circ \mathbf{x})}{dt}(t_0) = 0$, proving the statement.

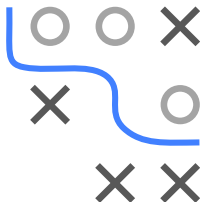
IMPORTANT PROPERTIES OF THE GRADIENT

- **Proof of 2.:** Let $\mathbf{v}$ be a vector with $\|\mathbf{v}\| = 1$ and θ the angle between $\mathbf{v}$ and $\nabla f(\mathbf{x})$.

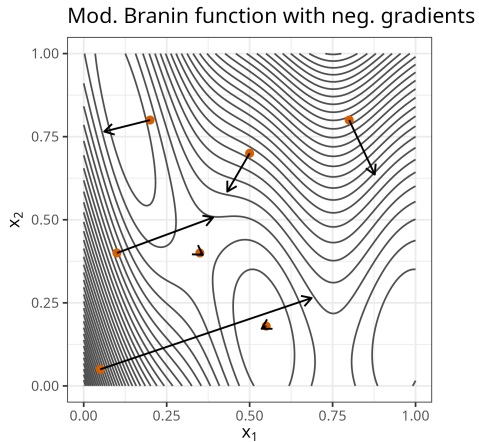
$$D_{\mathbf{v}}f(\mathbf{x}) = \nabla f(\mathbf{x})\mathbf{v} = \|\nabla f(\mathbf{x})\| \|\mathbf{v}\| \cos(\theta) = \|\nabla f(\mathbf{x})\| \cos(\theta)$$

by the cosine formula for dot products and $\|\mathbf{v}\| = 1$.

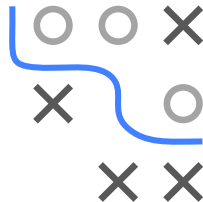
- $\cos(\theta)$ is maximal if $\theta = 0$, hence if $\mathbf{v}$ and $\nabla f(\mathbf{x})$ point in the same direction.
- (Alternative proof: Apply Cauchy-Schwarz to $\nabla f(\mathbf{x})\mathbf{v}$ and look for equality.)
- Analogous: Negative gradient $-\nabla f(\mathbf{x})$ points in direction of greatest *decrease*



IMPORTANT PROPERTIES OF THE GRADIENT



Length of arrows is norm of their gradient.



JACOBIAN MATRIX

For vector-valued

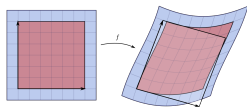
function $f : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}^m, \mathbf{x} \mapsto (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))^T$,

$f_j : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$, the **Jacobian** matrix $J_f : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}^{m \times d}$ generalizes gradient by placing all ∇f_j in its rows:

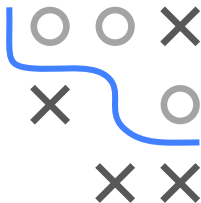
$$J_f(\mathbf{x}) \text{ or } \nabla f(\mathbf{x}) = \begin{pmatrix} \nabla f_1(\mathbf{x}) \\ \vdots \\ \nabla f_m(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x})}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_d} \end{pmatrix}$$

We will mainly use the ∇f notation.

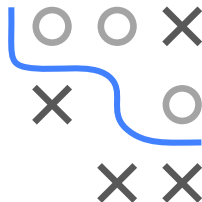
- Jacobian gives best linear approximation of distorted volumes



► [Click for source](#)



JACOBIAN DETERMINANT



- Let $f \in \mathcal{C}^1$ and $\mathbf{x}_0 \in \mathcal{S} \subseteq \mathbb{R}^d$.
- **Inverse function theorem:** Let $\mathbf{y}_0 = f(\mathbf{x}_0)$. If $\det(J_f(\mathbf{x}_0)) \neq 0$, then
 - 1 f is invertible in a neighborhood of $\mathbf{x}_0$,
 - 2 $f^{-1} \in \mathcal{C}^1$ with $J_{f^{-1}}(\mathbf{y}_0) = J_f(\mathbf{x}_0)^{-1}$.
- $|\det(J_f(\mathbf{x}_0))|$: factor by which f expands/shrinks volumes near $\mathbf{x}_0$
- If $\det(J_f(\mathbf{x}_0)) > 0$, f preserves orientation near $\mathbf{x}_0$
- If $\det(J_f(\mathbf{x}_0)) < 0$, f reverses orientation near $\mathbf{x}_0$

HESSIAN MATRIX

For real-valued function $f : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$, the **Hessian** matrix $\nabla^2 : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ contains all its second derivatives (if they exist):

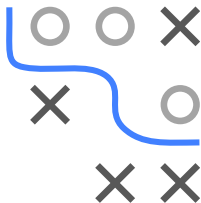
$$\nabla^2 f(\mathbf{x}) = \left(\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} \right)_{i,j=1,\dots,d}$$

Note: Hessian of f is Jacobian of ∇f . Also, the Hessian is often denoted by $H(\mathbf{x}) \triangleq \nabla^2 f(\mathbf{x})$. **Example:** Let $f(\mathbf{x}) = \sin(x_1) \cdot \cos(2x_2)$.

Then:

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} -\cos(2x_2) \cdot \sin(x_1) & -2 \cos(x_1) \cdot \sin(2x_2) \\ -2 \cos(x_1) \cdot \sin(2x_2) & -4 \cos(2x_2) \cdot \sin(x_1) \end{pmatrix}$$

- If $f \in \mathcal{C}^2$, then $\nabla^2 f$ is symmetric
- Many local properties (geometry, convexity, critical points) are encoded by the Hessian and its spectrum ($\rightarrow$ later)



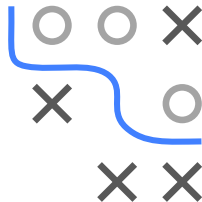
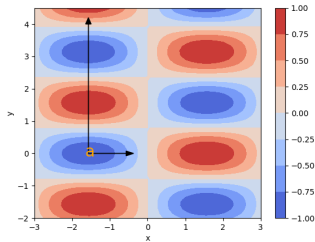
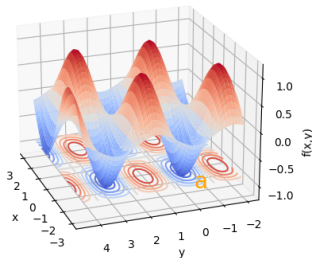
LOCAL CURVATURE BY HESSIAN

Eigenvector corresponding to largest (resp. smallest) **eigenvalue** of Hessian points in direction of largest (resp. smallest) **curvature**

Example (previous slide): For $\mathbf{a} = (-\pi/2, 0)^T$, we have

$$\nabla^2 f(\mathbf{a}) = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

and thus $\lambda_1 = 4, \lambda_2 = 1, \mathbf{v}_1 = (0, 1)^T$, and $\mathbf{v}_2 = (1, 0)^T$.



LIPSCHITZ CONTINUITY

Function $h : \mathcal{S} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}^m$ is **Lipschitz continuous** if slopes are bounded:

$$\|h(\mathbf{x}) - h(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \text{for each } \mathbf{x}, \mathbf{y} \in \mathcal{S} \subseteq \mathbb{R}^d \text{ and some } L > 0$$

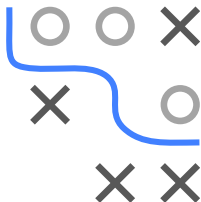
- **Examples** ($d = m = 1$): $\sin(x)$, $|x|$
- **Not** examples: $1/x$ (but *locally* Lipschitz continuous), $\sqrt{x}$
- If $m = d$ and h **differentiable**:

$$h \text{ Lipschitz continuous with constant } L \iff J_h \preceq L \cdot I_d$$

Note: $\mathbf{A} \preceq \mathbf{B} : \iff \mathbf{B} - \mathbf{A}$ is positive semidefinite, i.e.,
 $\mathbf{v}^T (\mathbf{B} - \mathbf{A}) \mathbf{v} \geq 0 \quad \forall \mathbf{v} \neq 0$. **Proof** of “ $\Rightarrow$ ” for $d = m = 1$:

$$h'(x) = \lim_{\epsilon \rightarrow 0} \frac{h(x + \epsilon) - h(x)}{\epsilon} \leq \lim_{\epsilon \rightarrow 0} \left| \frac{h(x + \epsilon) - h(x)}{\epsilon} \right| \leq \lim_{\epsilon \rightarrow 0} L = L$$

[**Proof** of “ $\Leftarrow$ ” by mean value theorem: Show that $\lambda_{\max}(J_h) \leq L$.]



LIPSCHITZ GRADIENTS / L-SMOOTHNESS

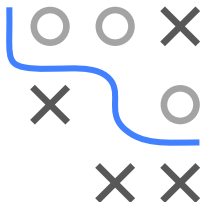
- f is called **L -smooth** (or has **L -Lipschitz gradients**) if

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{S}$$

- For $f \in \mathcal{C}^2$, since $\nabla^2 f$ is the Jacobian of ∇f and symmetric:

$$f \text{ is } L\text{-smooth} \iff -L\mathbf{I}_d \preceq \nabla^2 f(\mathbf{x}) \preceq L \cdot \mathbf{I}_d \quad \forall \mathbf{x}$$

- Equivalently, all eigenvalues of $\nabla^2 f$ have magnitude bounded by L
- **Interpretation:** Curvature in any direction is bounded by L
- Lipschitz gradients occur frequently in machine learning $\implies$
Fairly **weak assumption**



DESCENT LEMMA

- For L -smooth f : integrating the Lipschitz bound on ∇f yields a **quadratic upper bound**:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad \forall \mathbf{x}, \mathbf{y}$$

- Geometric reading: f is upper-bounded by a quadratic that touches f at $\mathbf{x}$
- Cornerstone for analyzing **gradient descent** (later: pick step size $1/L$, guaranteed progress per step)
- Counterpart of the lower bound for strongly convex f (see slide-set “Convexity”)

